

Smart Logistics Performance & Delay Analytics

End-to-end data analysis and Power BI dashboard case study

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1,000 shipment records · FY2024

Project objective: analyze logistics delay patterns, identify high-risk operational segments, validate important relationships statistically, and communicate the findings through an executive Power BI dashboard.

Technology: Python Pandas NumPy Matplotlib / Seaborn SciPy Power BI DAX

1 EXECUTIVE SUMMARY

RECORDS ANALYZED 1,000	DELAYED RECORDS 566	OVERALL DELAY RATE 56.60%	AVG. ASSET UTILIZATION 79.60%
AVG. WAITING TIME 35.06 min	AVG. INVENTORY LEVEL 297.92	AVG. TRANSACTION AMOUNT \$299.06	

2 KEY BUSINESS FINDINGS

Heavy traffic is the highest-risk segment

It recorded a **100% observed delay rate** in this dataset — the single sharpest finding in the analysis.

Traffic condition matters, statistically

A statistically significant association was found between Traffic_Status and Logistics_Delay. Cramer's V was **0.6104**, indicating a strong association.

Shipment status is strongly associated with delay

The notebook reports a highly significant chi-square result ($p \approx 1.47 \times 10^{-90}$).

Waiting time: significant, but the effect is small

The delayed group averaged **34.26** versus **36.10** for non-delayed records. The difference was statistically significant at the 5% level ($p \approx 0.0454$), but is small in practical terms.

Numeric variables alone are weak predictors

Correlation analysis showed only weak linear relationships with the delay flag — making categorical and operational segmentation (traffic, status) far more informative for this dataset than continuous sensor/business variables.

3 ANALYSIS METHODOLOGY

Stage	Work performed
Data quality	Shape, data types, missing values and duplicate checks.
Feature preparation	Timestamp/date/time features and delay-related helper fields.
Exploratory analysis	Shipment, traffic, asset, hourly and day-of-week segmentation.
Delay analysis	Delay rates, cross-tabulations, delay reasons and operational comparisons.
Statistical validation	Chi-square tests, Cramer's V and a two-sample t-test for waiting time.
Visualization	Seven core analytical charts plus executive Power BI visuals.
Reporting	KPI cards, slicers, reset control, operational charts and dashboard layout.

4 POWER BI DASHBOARD

The dashboard is designed as an executive operational view. It combines KPI cards with slicers and visual analysis of shipment status, traffic condition, overall delay distribution, day-wise delay rate, asset-wise delay rate and hourly delay patterns.



Fig. 1 — Smart Logistics Performance & Delay Analytics, Power BI Desktop

Dashboard interaction: slicers allow filtering by Shipment Status, Day Name, Delay Status, Asset ID and Traffic Status. The reset control returns the page to its default, unfiltered state.

5 PORTFOLIO DELIVERABLES

- ✓ Python exploratory data analysis notebook with documented cells and statistical validation.
- ✓ Power BI executive dashboard (.pbix).
- ✓ Cleaned / analysis-ready data workbook.
- ✓ GitHub-ready project structure with README, requirements and data dictionary.
- ✓ This business-focused project report for portfolio presentation.

6 RECOMMENDATIONS

- 1 Prioritize monitoring and intervention for heavy-traffic conditions.
- 2 Use shipment-status and traffic-status segmentation as operational alert dimensions.
- 3 Track asset-level delay rates to identify consistently high-risk trucks.
- 4 Monitor hourly and day-wise patterns when scheduling or allocating capacity.
- 5 Validate these patterns on a larger, independently collected operational dataset before using them for production decisions.

7 LIMITATIONS

The findings describe relationships observed in this dataset and do not establish causation. The heavy-traffic 100% rate is an observed sample result, not a universal probability. The dataset contains 1,000 records, so external validation and larger samples would strengthen confidence before operational deployment.

8 PORTFOLIO POSITIONING

This project demonstrates an end-to-end analytics workflow: raw operational data → quality checks → feature analysis → statistical validation → business KPIs → Power BI dashboard → documented portfolio deliverables.

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